

Aiport Data analysis

Problem statement 1:

Gets Ideas of Tarvel insights

-----> Between two pair of airports

-----> Help me to find

1. Most frequent route

2. Find out No of passengers details based on that -----> I can take decisions to plan som airline operations

Q.1.find Total No of passengers travel between which 2 airports?

Ans : **select**

Origin_airport,

destination_airport,

sum(passengers) as Total_passenegers

from airport2

group by

Origin_airport,

destination_airport

order by Total_passenegers desc;



The screenshot shows a 'Result Grid' with a toolbar at the top containing icons for 'Filter Rows', 'Export', and 'Wrap Cell Content'. The table below displays the top 10 flight routes, sorted by the total number of passengers in descending order. Each row is highlighted with a light blue background.

	Origin_airport	destination_airport	Total_passenegers
▶	JAI	BLR	46175
	PNQ	DEL	45100
	GOI	BOM	44025
	AMD	JAI	42950
	CCU	PNQ	41875
	MAA	GOI	40800
	HYD	AMD	39725
	BLR	CCU	38650
	DEL	MAA	37575

Q.2 Problem statment

1. Identify Highest & lowest Seat Occupancy

2. Optimize Flight Capacity

3. Improving Operational efficiency

Solution:-

select

Origin_airport,

destination_airport,

avg((passengers)/(seats))*100 as seats_available

from airport2

group by

Origin_airport,

destination_airport

order by

seats_available desc;

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Origin_airport	destination_airport	seats_available	
JAI	BLR	91.14800000	
PNQ	DEL	91.02400000	
GOI	BOM	90.46600000	
CCU	PNQ	89.66200000	
MAA	GOI	89.27000000	
BLR	CCU	89.06800000	
DEL	MAA	88.92200000	
AMD	JAI	88.73400000	
BOM	HYD	88.23200000	
HYD	AMD	88.07000000	

Q.3 Problem statment

Find out most frequent Travel route

-->optimize Resume collection

-->enhance the service

Solution:

select

Origin_airport,

destination_airport,

origin_city,

destination_city,

sum(passengers) as Total_passenegers

from airport2

group by

Origin_airport,

destination_airport,

origin_city,

destination_city

order by Total_passenegers desc

limit 5;

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Fetch rows:

	Origin_airport	destination_airport	origin_city	destination_city	Total_passenegers
▶	JAI	BLR	Jaipur	Bengaluru	46175
	PNQ	DEL	Pune	Delhi	45100
	GOI	BOM	Goa	Mumbai	44025
	AMD	JAI	Ahmedabad	Jaipur	42950
	CCU	PNQ	Kolkata	Pune	41875

Q.4 Problem statment

Find activity level at various origin cities

----> I can find out the key Hub

---->To take decisions for some flight operations & capacity management

Solution:

```
select
origin_city,
count(flights) as Total_flights,
sum(passengers) as Total_passengers
from
airport2
group by
origin_city
order by
Total_passengers desc;
```

origin_city	Total_flights	Total_passengers
Jaipur	5	46175
Pune	5	45100
Goa	5	44025
Ahmedabad	5	42950
Kolkata	5	41875
Chennai	5	40800
Hyderabad	5	39725
Bengaluru	5	38650
Delhi	5	37575
Mumbai	5	36500

Q.5 Problem statment

Look into travel patterns

Future route Planning

Solution :-

select

origin_city,

origin_airport,

sum(distance) as Total_distance

from airport2

group by origin_city,origin_airport

order by Total_distance desc;

	origin_city	origin_airport	Total_distance
▶	Goa	GOI	6445
	Delhi	DEL	6135
	Jaipur	JAI	5915
	Ahmedabad	AMD	5760
	Hyderabad	HYD	5605
	Mumbai	BOM	5450
	Pune	PNQ	5230
	Kolkata	CCU	5075
	Bengaluru	BLR	4920
	Chennai	MAA	4390

Q.6 Problem statement

Find out seasonal trends (in which month most of passengers travelled)

How----> no of flights

No of passenger

Solution :-

select

Year(fly_date) as Year,

Month(fly_date) as Month,

count(flights) as Total_flights,

sum(passengers) as Total_passengers,

avg(distance) as Avg_distance

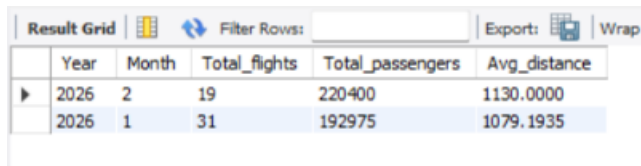
from airport2

group by

Year,Month

order by

Year desc,Month desc;



The screenshot shows a 'Result Grid' with columns: Year, Month, Total_flights, Total_passengers, and Avg_distance. The data is filtered for the year 2026. The first row shows Month 2 with 19 flights and 220,400 passengers, resulting in an average distance of 1130.0000. The second row shows Month 1 with 31 flights and 192,975 passengers, resulting in an average distance of 1079.1935.

	Year	Month	Total_flights	Total_passengers	Avg_distance
▶	2026	2	19	220400	1130.0000
	2026	1	31	192975	1079.1935

Q.7 Problem statement

Find the underutilized route (least used routes)

Want to make proper capacity management

Want to adjust route adjustment

How -----> find passengers seats to ratio for origin & destination

Formula

= passengers seats (total no of passengers ÷ total no of seats)

= passengers < 0.5

Solution

select

origin_airport,

destination_airport,

sum(passengers) as total_passengers,

sum(seats) as total_seats,

sum(passengers)*1.0/sum(seats) as passengers_seats

```

from airport2
group by
origin_airport,
destination_airport
having
passengers_seats < 10000
order by
passengers_seats ;

```

Result Grid					
		Filter Rows:	Export:	Wrap Cell Content:	
	origin_airport	destination_airport	total_passengers	total_seats	passengers_seats
	BOM	HYD	36500	40920	0.89198
	HYD	AMD	39725	44265	0.89744
	DEL	MAA	37575	41755	0.89989
	AMD	JAI	42950	47610	0.90212
▶	MAA	GOI	40800	45100	0.90466
	BLR	CCU	38650	42590	0.90749
	GOI	BOM	44025	48445	0.90876
	CCU	PNQ	41875	45935	0.91161
	PNQ	DEL	45100	49280	0.91518
	JAI	BLR	46175	50115	0.92138

Q.8 Problem statement

Find active airport

Find out the airport where highest frequency of Flights

Benefits :- airline & stakeholder can optimize flgths scheduling & they can also improve services

Solution :-

```

select origin_airport,origin_city,
sum(flights) as total_flights
from airport2

```

group by origin_airport,origin_city

order by total_flights desc;

	origin_airport	origin_city	total_flights
▶	MAA	Chennai	225
	HYD	Hyderabad	210
	JAI	Jaipur	210
	BLR	Bengaluru	195
	PNQ	Pune	195
	DEL	Delhi	180
	GOI	Goa	180
	BOM	Mumbai	165
	AMD	Ahmedabad	165
	CCU	Kolkata	150

Q.9 Problem statement

Find a city where most of passengers travels

Formula using :- Exclusive OR <> "Mumbai"

Solution:-

select

origin_city,

destination_city,

sum(flights) as total_flights,

sum(passengers) as total_passengers

from airport2

where

destination_city = "Mumbai" and

origin_city<>"Mumbai"

group by

origin_city,destination_city

order by

total_flights ,

total_passengers desc;

	origin_city	total_flights	total_passengers
▶	Goa	180	44025

	origin_city	total_flights	total_passengers
▶	Goa	180	44025

Q.10 Problem statement

Find Maximum extensive travel connection

Flights which travel maximum distance

Solution:-

select

origin_airport,

destination_airport,

Max(distance) as Long_distance

from

airport2

group by

origin_airport,

destination_airport

order by

Long_distance desc;

	origin_airport	destination_airport	Long_distance
▶	DEL	MAA	2067
	GOI	BOM	2049
	HYD	AMD	2031
	BOM	HYD	1930
	AMD	JAI	1912
	BLR	CCU	1894
	JAI	BLR	1793
	CCU	PNQ	1775
	PNQ	DEL	1656
	MAA	GOI	1638

11.Problem statement (i created my own problem statement)

Calculate the origincity ,destination city ,total passenegers ,total flights, total seats , total seats , find out the dates of all flights

Between Two cities

Hint:-(mumbai to hyderabad and goa to mumbai)

Solution

select

origin_city,

destination_city,

fly_date as all_dates,

sum(passengers) as total_passengers,

sum(flights) as total_flights,

sum(seats) as total_seats,

sum(distance) as total_distance,

count(fly_date) as fly_days

from

airport2

where

origin_city in ('Mumbai','Goa')

and destination_city in (

'Delhi',

'Goa',

'Chennai',

'Hyderabad',

'Mumbai'

)

group by

origin_city,

destination_city,

fly_date;

	origin_city	destination_city	all_dates	total_passengers	total_flights	total_seats	total_distance	fly_days
▶	Mumbai	Hyderabad	2026-01-01	3000	18	3500	250	1
	Goa	Mumbai	2026-01-08	4505	39	5005	1209	1
	Mumbai	Hyderabad	2026-01-11	5150	48	6010	1620	1
	Goa	Mumbai	2026-01-18	6655	24	7515	679	1
	Mumbai	Hyderabad	2026-01-21	7300	33	8520	1090	1
	Goa	Mumbai	2026-01-28	8805	54	10025	2049	1
	Mumbai	Hyderabad	2026-01-31	9450	18	10190	560	1
	Goa	Mumbai	2026-02-07	10955	39	11695	1519	1
	Mumbai	Hyderabad	2026-02-10	11600	48	12700	1930	1
	Goa	Mumbai	2026-02-17	13105	24	14205	989	1

12.Problem statement

**** Insights on seasonal trends in travel**

-----> analysis – Find out most & least Count of flight across multiple years

How ----> Find count of flights based on month

Solution:-

with Monthly_flights as (select

Month(fly_date) as Month,

```
sum(flights) as Total_flights
```

```
from airport2
```

```
group by
```

```
Month(fly_date) )
```

```
select
```

```
Month,
```

```
Total_flights,
```

```
CASE
```

```
when Total_flights = (select MAX(Total_flights) from Monthly_flights ) then 'Most Busy'
```

```
when Total_flights = (select Min(Total_flights) from Monthly_flights ) then 'least Busy'
```

```
Else NULL
```

```
END as status
```

```
from Monthly_flights
```

```
Where
```

```
Total_flights = (select MAX(Total_flights) from Monthly_flights ) or
```

```
Total_flights = (select Min(Total_flights) from Monthly_flights );
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
	Month	Total_flights	status
▶	1	4752	Most Busy
	2	2724	least Busy

13.Problem statement

Calculate year over year percentage growth in passengers ----->origin----->destination

1.passenger summary

2.passenger growth

Solution :-

with passenger_summary as

(select

origin_airport,

destination_airport,

sum(passengers) as total_passengers,

Year(fly_date) as Year

from airport2

group by

origin_airport,

destination_airport,

Year(fly_date)),

passenger_growth as

(select

origin_airport,

destination_airport,

year,

total_passengers,

lag(total_passengers) over (partition by origin_airport ,destination_airport order by year) as
previous_year_passenger

from

passenger_summary)

select

origin_airport,

destination_airport,

```

year,
total_passengers,
case
when previous_year_passenger is not null then
((total_passengers-previous_year_passenger)*100/nullif(previous_year_passenger,0))
End as Growth_percentage
from
passenger_growth
order by
origin_airport,
destination_airport,
year;

```

	origin_airport	destination_airport	year	total_passengers	Growth_percentage
▶	AMD	JAI	2023	42950	NULL
	AMD	JAI	2024	42950	0.0000
	AMD	JAI	2025	42950	0.0000
	AMD	JAI	2026	42950	0.0000
	BLR	CCU	2023	38650	NULL
	BLR	CCU	2024	38650	0.0000
	BLR	CCU	2025	38650	0.0000
	BLR	CCU	2026	38650	0.0000
	BOM	HYD	2023	36500	NULL
	BOM	HYD	2024	36500	0.0000
	BOM	HYD	2025	36500	0.0000
	BOM	HYD	2026	36500	0.0000
	CCU	PNQ	2023	41875	NULL
	CCU	PNQ	2024	41875	0.0000
	CCU	PNQ	2025	41875	0.0000
	CCU	PNQ	2026	41875	0.0000
	DEL	MAA	2023	37575	NULL
	DEL	MAA	2024	37575	0.0000
	DEL	MAA	2025	37575	0.0000
	DEL	MAA	2026	37575	0.0000

14.problem statment

Finding Trending route

-----> consistent path

-----> Growth in percentage

How --->

- Based on origin- destination
- year to year growth (flights)

step 1. flights summary(count/sum)

Step.2 Flight growth(lag)

Step 3.Growth rate (find result)

Solution

1.Growth summary

	origin_airport	destination_airport	year	total_flights
►	BOM	HYD	2026	5
	DEL	MAA	2026	5
	BLR	CCU	2026	5
	HYD	AMD	2026	5
	MAA	GOI	2026	5
	CCU	PNQ	2026	5
	AMD	JAI	2026	5
	GOI	BOM	2026	4
	PNQ	DEL	2026	5
	JAI	BLR	2026	5
	BOM	HYD	2025	5
	DEL	MAA	2025	5
	BLR	CCU	2025	5
	HYD	AMD	2025	5
	MAA	GOI	2025	5
	CCU	PNQ	2025	5

2.Flight growth rate then previous year

	origin_airport	destination_airport	year	total_flights	pervious_year_flights
▶	AMD	JAI	2023	5	NULL
	AMD	JAI	2024	5	5
	AMD	JAI	2025	5	5
	AMD	JAI	2026	5	5
	BLR	CCU	2023	5	NULL
	BLR	CCU	2024	5	5
	BLR	CCU	2025	5	5
	BLR	CCU	2026	5	5
	BOM	HYD	2023	5	NULL
	BOM	HYD	2024	5	5
	BOM	HYD	2025	5	5
	BOM	HYD	2026	5	5
	CCU	PNQ	2023	5	NULL
	CCU	PNQ	2024	5	5
	CCU	PNQ	2025	5	5
	CCU	PNQ	2026	5	5

3. Growth indicator

	origin_airport	destination_airport	year	total_flights	growth_rate	growth_indicator
▶	AMD	JAI	2023	5	NULL	0
	AMD	JAI	2024	5	0.00000	0
	AMD	JAI	2025	5	0.00000	0
	AMD	JAI	2026	5	0.00000	0
	BLR	CCU	2023	5	NULL	0
	BLR	CCU	2024	5	0.00000	0
	BLR	CCU	2025	5	0.00000	0
	BLR	CCU	2026	5	0.00000	0
	BOM	HYD	2023	5	NULL	0
	BOM	HYD	2024	5	0.00000	0
	BOM	HYD	2025	5	0.00000	0
	BOM	HYD	2026	5	0.00000	0
	CCU	PNQ	2023	5	NULL	0
	CCU	PNQ	2024	5	0.00000	0
	CCU	PNQ	2025	5	0.00000	0
	CCU	PNQ	2026	5	0.00000	0
	DEL	MAA	2023	5	NULL	0
	DEL	MAA	2024	5	0.00000	0

4. Final Growth rate then previous showing with 1 and 0

origin_airport	destination_airport	maximum_growth_rate	minimum_growth_rate
AMD	JAI	0.00000	0.00000
BLR	CCU	0.00000	0.00000
BOM	HYD	0.00000	0.00000
CCU	PNQ	0.00000	0.00000
DEL	MAA	0.00000	0.00000
HYD	AMD	0.00000	0.00000
JAI	BLR	0.00000	0.00000
MAA	GOI	0.00000	0.00000
PNQ	DEL	0.00000	0.00000

Code:-

```
With growth_summary as(
select
origin_airport,
destination_airport,
Year(fly_date) as year,
count(flights) as total_flights
from
airport2
group by
origin_airport,
destination_airport,
year),
flight_growth as
(select
origin_airport,
destination_airport,
year,
total_flights,
lag(total_flights) over (partition by origin_airport,destination_airport order by year) as
previous_year_flights
from
growth_summary),
growth_rates as(
select
origin_airport,
```

```

destination_airport,
year,
total_flights,
case
when
(previous_year_flights is not null and previous_year_flights)>0 then
((total_flights-previous_year_flights)*100.0/previous_year_flights)
else
null
    End as growth_rate,
    case
when
(previous_year_flights is not null and previous_year_flights)>previous_year_flights
then
1
else 0
    End as growth_indicator
from
flight_growth)

select
origin_airport,
destination_airport,
Max(growth_rate) as maximum_growth_rate,
Min(growth_rate) as minimum_growth_rate
from growth_rates

```

where

growth_indicator=0

group by

origin_airport,

destination_airport

having

Min(growth_rate)=0

order by

origin_airport,

destination_airport;

15.Problem statement

*** Find out passenger to seat ratio Of Top 3 Airport**

(Based on Total number of flights (performance) operation efficient + flight volume

How ?

Find Top 3 airport (origin airport)

----->Highest weighted passenger to seat utilization ratio

-----> Considering to total number of flights)

Solution

1.utilization_ratio

	origin_airport	total_passengers	total_seats	total_flights	passenger_seat_ratio
►	BOM	146000	163680	20	0.89198
	DEL	150300	167020	20	0.89989
	BLR	154600	170360	20	0.90749

2.weighted_utilization

	origin_airport	total_seats	total_passengers	total_flights	passenger_seat_ratio	wighted_utilization
▶	BOM	163680	146000	20	0.89198	0.089646231
	DEL	167020	150300	20	0.89989	0.090441206
	BLR	170360	154600	20	0.90749	0.091205025

3. Final wighted_utilization

	origin_airport	total_seats	total_passengers	total_flights	wighted_utilization
▶	JAI	200460	184700	20	0.092601005
	PNQ	197120	180400	20	0.091977889
	CCU	183740	167500	20	0.091619095

16.Problem statement

Find out the seasonal travel pattern

Step 1.Based on specific city

How

Step 2.(*find the peak traffic month from each city with highest number of passengers)

→ 3 months passengers counts same

Step 3

1.monthly passengers count

2.max-passengers count based on city

3.peak months---->(if there is any tie then we will use self join)

Solution:-

1.monthly passengers count

	origin_city	year	month	total_passengers
►	Mumbai	2026	1	24900
	Delhi	2026	1	16095
	Bengaluru	2026	1	16740
	Hyderabad	2026	1	17385
	Chennai	2026	1	18030
	Kolkata	2026	1	18675
	Ahmedabad	2026	1	19320
	Goa	2026	1	19965
	Pune	2026	1	20610
	Jaipur	2026	1	21255
	Delhi	2026	2	21480
	Bengaluru	2026	2	21910
	Hyderabad	2026	2	22340
	Chennai	2026	2	22770
	Kolkata	2026	2	23200
	Ahmedabad	2026	2	23630

2.max-passengers count based on city

	origin_city	peak_passengers
►	Mumbai	24900
	Delhi	21480
	Bengaluru	21910
	Hyderabad	22340
	Chennai	22770
	Kolkata	23200
	Ahmedabad	23630
	Goa	24060
	Pune	24490
	Jaipur	24920

3.final output we got of peak passengers in each year,month from origin_city

	origin_city	year	month	total_passengers
▶	Ahmedabad	2023	2	23630
	Ahmedabad	2024	2	23630
	Ahmedabad	2025	2	23630
	Ahmedabad	2026	2	23630
	Bengaluru	2023	2	21910
	Bengaluru	2024	2	21910
	Bengaluru	2025	2	21910
	Bengaluru	2026	2	21910
	Chennai	2023	2	22770
	Chennai	2024	2	22770
	Chennai	2025	2	22770
	Chennai	2026	2	22770
	Delhi	2023	2	21480
	Delhi	2024	2	21480
	Delhi	2025	2	21480
	Delhi	2026	2	21480

17.problem statement

Find the routes which demand is reduced

identify the source to destination that they have experinced largest decimal in passenger year over year

How to find the solution

1. Yearly passenger count

2. Yearly decline

|
|
|

3. Percentage

Solution

WITH Yearly_passenger_count AS (

SELECT

origin_airport,

destination_airport,

```

    YEAR(fly_date) AS year,
    SUM(passengers) AS total_passenger
FROM airport2
GROUP BY
    origin_airport,
    destination_airport,
    YEAR(fly_date)
),

```

```

yearly_decline AS (
SELECT
    y1.origin_airport,
    y1.destination_airport,
    y1.year AS year1,
    y1.total_passenger AS passengers_year1,
    y2.year AS year2,
    y2.total_passenger AS passengers_year2,
    (
        (y2.total_passenger - y1.total_passenger)
        / NULLIF(y1.total_passenger, 0)
    ) * 100 AS percentage_change
FROM Yearly_passenger_count y1
JOIN Yearly_passenger_count y2
    ON y1.origin_airport = y2.origin_airport
    AND y1.destination_airport = y2.destination_airport
    AND y2.year = y1.year + 1

```

)

SELECT

origin_airport,

destination_airport,

year1,

passengers_year1,

year2,

passengers_year2,

percentage_change

FROM yearly_decline

WHERE percentage_change < 0

ORDER BY percentage_change

LIMIT 5;

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	origin_airport	destination_airport	year1	passengers_year1	year2	passengers_year2	percentage_change
▶	GOI	BOM	2025	44025	2026	30920	-29.7672

Problem statement 18

----->Find the underperforming routes

----->origin -----> destination

Solution:-

1.

	origin_airport	destination_airport	total_flights	total_passengers	total_seats	avg_seat_utilization
►	BOM	HYD	20	146000	163680	0.8920
	DEL	MAA	20	150300	167020	0.8999
	BLR	CCU	20	154600	170360	0.9075
	HYD	AMD	20	158900	177060	0.8974
	MAA	GOI	20	163200	180400	0.9047
	CCU	PNQ	20	167500	183740	0.9116
	AMD	JAI	20	171800	190440	0.9021
	GOI	BOM	19	162995	179575	0.9077
	PNQ	DEL	20	180400	197120	0.9152
	JAI	BLR	20	184700	200460	0.9214

2.

	origin_airport	destination_airport	total_flights	total_passengers	total_seats	avg_seat_utilization_percentage
►	BOM	HYD	20	146000	163680	89.20
	HYD	AMD	20	158900	177060	89.74
	DEL	MAA	20	150300	167020	89.99
	AMD	JAI	20	171800	190440	90.21
	MAA	GOI	20	163200	180400	90.47
	BLR	CCU	20	154600	170360	90.75
	GOI	BOM	19	162995	179575	90.77
	CCU	PNQ	20	167500	183740	91.16
	PNQ	DEL	20	180400	197120	91.52
	JAI	BLR	20	184700	200460	92.14

WITH flights_stats AS (

SELECT

origin_airport,

destination_airport,

COUNT(flights) AS total_flights,

SUM(passengers) AS total_passengers,

SUM(seats) AS total_seats,

```
SUM(passengers) / NULLIF(SUM(seats), 0) AS avg_seat_utilization
FROM airport2
GROUP BY origin_airport, destination_airport
)

SELECT
    origin_airport,
    destination_airport,
    total_flights,
    total_passengers,
    total_seats,
    ROUND(avg_seat_utilization * 100, 2) AS avg_seat_utilization_percentage
FROM flights_stats
WHERE total_flights >= 10
ORDER BY avg_seat_utilization_percentage;
```

Problem statement 19

----->Longest avg distance route

-----> Insights to the airlines for long haul travel pattern

Solution

1.

	origin_airport	destination_airport	avg_flight_distance
▶	BOM	HYD	1090.0000
	DEL	MAA	1227.0000
	BLR	CCU	984.0000
	HYD	AMD	1121.0000
	MAA	GOI	878.0000
	CCU	PNQ	1015.0000
	AMD	JAI	1152.0000
	GOI	BOM	1304.7895
	PNQ	DEL	1046.0000
	JAI	BLR	1183.0000

2.

	origin_airport	destination_airport	avg_flight_distance
▶	GOI	BOM	1304.79
	DEL	MAA	1227.00
	JAI	BLR	1183.00
	AMD	JAI	1152.00
	HYD	AMD	1121.00
	BOM	HYD	1090.00
	PNQ	DEL	1046.00
	CCU	PNQ	1015.00
	BLR	CCU	984.00
	MAA	GOI	878.00

